Submission Summary

Conference Name

25th Mexican International Conference on Artiticial Intelligence

Paper ID

38

Paper Title

A Two-Condition Diagnostic for Decoder-Based Multi-Objective Search: When Blind Sampling Beats a Mismatched NSGA-II

Abstract

Many hard-constrained combinatorial problems are solved by pairing a metaheuristic with a feasibility-preserving decoder, yet the true driver of performance is seldom isolated. We isolate it on a calibrated synthetic case study: allocating 140,000 U.S. employment-based visas per year under a per-country ceiling and spillover cascade—a three-objective integer program with six hard constraints, searched through a random-key decoder whose allocations are feasible by construction—no penalties, no repair. A controlled nine-method ladder shows why search fails. Blind random sampling beats a standard real-coded NSGA-II and a naive swarm because their realized search is degenerate: interpolating operators barely change the decoded order, or gated acceptance freezes the population. The two corresponding conditions are necessary there, not sufficient: a BRKGA-style order-changing crossover under diversity-preserving selection only ties blind sampling, and a registered disruption sweep finds hypervolume nearly flat within that family—only the right kind of order renewal reaches the top tier. We package these measurements as a cheap a priori screening diagnostic and stress it with registered prospective tests: the two-condition core predicted correctly; the landscape label did not. Across knapsack, traveling-salesman, flow-shop, and set-covering replications, the best ladder method satisfies both conditions, while the mismatched genetic algorithm's collapse below blind sampling is landscape- and configuration-specific. A Taguchi design fixes shared parameters; the recovered front Pareto-dominates the first-in, first-out incumbent.

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Primary Subject Area

Genetic Algorithms

Secondary Subject Areas

Hybrid Intelligent Systems
Planning and Scheduling

Submission Files

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Submission Questions Response

1. First author

First author is student?

Yes, master